Feasible-First Exploration for Constrained ML Deployment Optimization in Crash-Prone Hierarchical Search Spaces

Christian Lysenstøen
Department of Electrical Engineering and Computer Sciences
University of California, Berkeley
christian@lysenstoen.net

March 2026

Abstract

Deploying machine learning models under production constraints requires joint optimization over model family, quantization scheme, runtime backend, and serving configuration. This induces a hierarchical mixed-variable search space in which many configurations are invalid: evaluations may crash, exceed memory limits, or violate latency constraints. Standard black-box optimizers such as Tree-structured Parzen Estimators (TPE) and constrained Bayesian optimization are effective when valid configurations are common, but they can spend a large fraction of a small evaluation budget on invalid or uninformative trials in hostile deployment spaces. This paper studies that regime and asks whether optimization should be decomposed into an explicit exploration stage followed by model-guided exploitation.

We propose *Thermal Budget Annealing* (TBA), a feasible-first exploration procedure that maps valid and feasible regions before warm-starting TPE. We also introduce *DeployBench*, a benchmark suite for deployment optimization with hierarchical structure, hidden crash zones, hard constraints, and unequal evaluation costs. On synthetic benchmarks and real GPU deployment with five pre-trained vision models on an NVIDIA RTX 5080 Laptop GPU, the proposed hybrid improves model-family discovery under tight constraints while reducing wasted budget relative to cold-start TPE. The results support a narrower but practically important conclusion: in partially invalid deployment spaces, early exploration quality determines which model families are available to the exploitation phase.

Keywords: black-box optimization, constrained optimization, deployment optimization, warm-start optimization, hierarchical search, hidden constraints, crash-aware search, AutoML

1 Introduction

Modern ML deployment rarely consists of a single model-selection decision. Practical deployment requires choosing a model family, quantization scheme, runtime backend, batch size, and backend-specific execution parameters under strict constraints on latency, memory, and, in some settings, power. The resulting optimization problem is structurally different from conventional hyperparameter optimization. First, the search space is *hierarchical*: the activeness of some parameters depends on earlier structural choices. Second, the space is often *partially invalid*: some configurations fail outright because of out-of-memory events, unsupported backend–operator combinations, compiler failures, or quantization incompatibilities. Third, evaluation costs are unequal because exporting, compiling, and benchmarking different configurations can vary from seconds to minutes.

These characteristics matter most when the evaluation budget is small. A method that spends its first several trials on invalid configurations may never observe the model family that would have produced the best feasible deployment. This issue is especially acute for model-guided optimizers that rely on early samples to shape later exploitation. If early observations are narrow, then later sampling may reinforce that narrow view. The central question in this paper is therefore not whether a new optimizer can dominate established methods in all settings, but whether a *feasible-first exploration stage* improves performance in deployment spaces that are crash-heavy, hierarchical, and budget-constrained.

This question is timely for two reasons. First, recent work in hyperparameter optimization and constrained optimization has emphasized hierarchical spaces, hidden constraints, and warm-starting as first-class concerns rather than secondary engineering details [Akiba et al., 2019, Awad et al., 2021, Lindauer et al., 2022, Watanabe and Hutter, 2023, Tian et al., 2024]. Second, deployment optimization itself has become a systems problem: practical inference quality depends on the interaction between model architecture, quantization, runtime, and hardware target, not on model choice alone [Chen et al., 2018, Reddi et al., 2020, Zaharia et al., 2024, Agrawal et al., 2024].

The proposed method is simple. Phase 1 uses crash-aware simulated annealing to search for feasible regions and diversify model-family discovery. Phase 2 warm-starts Optuna's TPE sampler from the Phase 1 history. The resulting method, TBA→TPE, is not presented as a replacement for TPE in benign spaces. The claim is narrower: when invalidity is high and budgets are small, explicit feasible-first exploration can improve model-family discovery and reduce wasted budget enough to make model-guided exploitation more effective.

Contributions.

1. We formalize budget-constrained deployment optimization in partially invalid hierarchical spaces and identify *premature exploitation* as a central failure mode of model-guided search in crash-heavy regimes.

2. We propose TBA, a feasible-first exploration procedure with adaptive temperature control, crash-aware state transitions, and an adaptive handoff into TPE.

3. We introduce *DeployBench*, a benchmark suite for constrained deployment search with hidden crash regions, hierarchical conditionals, and unequal evaluation costs.

4. We show on synthetic tasks and real GPU deployment that warm-starting TPE with feasible-first exploration improves discovery of the best model family under tight constraints while reducing wasted budget relative to cold-start TPE.

2 Background and Problem Setting

Deployment optimization differs from conventional hyperparameter optimization in ways that are operationally significant. In training-time HPO, a trial typically returns a scalar objective together with optional soft constraints [Hutter et al., 2019]. In deployment optimization, an evaluation may fail to return any useful objective at all due to runtime incompatibilities, export failures, out-of-memory conditions, or compiler exceptions. This creates a *partially invalid* search problem in which optimization must reason simultaneously about objective quality, constraint satisfaction, and evaluability.

A second distinction is that deployment search spaces are naturally *hierarchical*. Structural decisions such as backend selection activate or deactivate lower-level parameters. For example, selecting `torch_compile` introduces compiler-related options that are irrelevant under eager execution, while selecting `onnxruntime` activates export and execution-provider choices. Consequently, the dimensionality of the active search space depends on the configuration itself.

A third distinction is that deployment trials have highly unequal cost. A configuration that runs in eager mode on a small model may complete in seconds, whereas a configuration that triggers graph export, compilation, or repeated failed allocation attempts may consume substantially more wall-clock time before yielding either a result or an exception. This cost asymmetry means that wasted evaluations are not merely statistically inefficient; they are operationally expensive.

These observations motivate a decomposition of the optimization process into two conceptual tasks: *region discovery* and *within-region optimization*. Region discovery concerns identifying model families and system configurations that can be evaluated successfully and satisfy constraints. Within-region optimization concerns improving the deployment objective after such regions have been identified. The central premise of this work is that standard model-guided optimizers conflate these tasks too early in crash-heavy settings.

3 Related Work

3.1 Hyperparameter Optimization

The field of hyperparameter optimization has matured substantially over the past decade. Random search [Bergstra and Bengio, 2012] established that uniform sampling can outperform grid search in high-dimensional spaces. Bayesian optimization with Gaussian processes [Snoek et al., 2012, Garnett, 2023] introduced model-guided sequential search, and Tree-structured Parzen Estimators [Bergstra et al., 2011] provided a scalable alternative for conditional spaces. Multi-fidelity methods such as Hyperband [Li et al., 2017] and BOHB [Falkner et al., 2018] reduce cost by truncating unpromising trials, while DEHB [Awad et al., 2021] combines Hyperband with evolutionary search for improved scalability. Optuna [Akiba et al., 2019] and SMAC3 [Lindauer et al., 2022] provide practical frameworks with support for conditional search spaces, and the recent c-TPE extension [Watanabe and Hutter, 2023] adds native constraint-aware sampling to the TPE family. Existing HPO benchmarks [Eggensperger et al., 2021, Pfisterer et al., 2022] have standardized evaluation methodology but generally assume that most configurations are evaluable.

These methods are designed for spaces where the vast majority of configurations produce valid results. The deployment setting studied here, in which 30–80% of random configurations may crash or violate constraints, lies outside those assumptions.

3.2 Constrained and Hidden-Constraint Optimization

Constrained Bayesian optimization has been studied through probability-of-feasibility weighting [Gardner et al., 2014], unknown-constraint modeling [Gelbart et al., 2014], and information-based acquisition strategies [Hernández-Lobato et al., 2016]. More recently, optimization under hidden or partially observed constraints has received renewed attention [Tian et al., 2024]. The present work connects that perspective to ML deployment, where crashes are not merely noisy evaluations but an intrinsic feature of the search space.

3.3 ML Systems Optimization

Deployment optimization for ML systems has gained attention through compiler-level approaches such as TVM [Chen et al., 2018] and Ansor [Zheng et al., 2020], which automate operator scheduling. TensorRT and ONNX Runtime provide backend-specific runtime optimization. MLPerf [Reddi et al., 2020] has established the importance of inference benchmarking under realistic latency constraints. More recent systems work has examined edge deployment [Hymel et al., 2023], large-scale inference simulation [Agrawal et al., 2024], and the broader shift toward compound AI systems where deployment configuration spans multiple interacting components [Zaharia et al.,

2024]. The problem considered here differs from this literature by treating deployment configuration itself as a constrained search problem over model family, quantization mode, backend, and serving parameters.

3.4 Simulated Annealing and Hybrid Methods

Simulated annealing [Kirkpatrick et al., 1983] is a classical metaheuristic for combinatorial optimization. Adaptive temperature schedules and reheating strategies are well studied [Ingber, 1989]. Evolutionary strategies [Hansen and Ostermeier, 2001] provide an alternative paradigm for black-box search in mixed spaces. The contribution of the present work is not a new simulated annealing theory, but the use of feasible-first annealing as an exploration front-end for model-guided optimization in crash-heavy deployment spaces.

4 Problem Formalization

4.1 Hierarchical Deployment Search Space

Definition 4.1 (Hierarchical mixed-variable search space)**.** *A hierarchical mixed-variable search space $\mathcal{X}$ is defined by variables $\{x_1, \ldots, x_d\}$ where each variable has a type $\tau_i \in \{\text{categorical}, \text{integer}, \text{continuous}\}$, a domain $\mathcal{D}_i$, and an optional activation condition $\phi_i : \mathcal{X} \to \{0, 1\}$. A configuration $\boldsymbol{x} \in \mathcal{X}$ assigns values only to active variables. The active set $\mathcal{A}(\boldsymbol{x}) = \{i : \phi_i(\boldsymbol{x}) = 1\}$ depends on the configuration itself.*

In the deployment setting studied here,

$$\mathcal{X} = \mathcal{M} \times \mathcal{Q} \times \mathcal{B} \times \mathcal{S} \times \prod_{i \in \mathcal{A}(\boldsymbol{x})} \mathcal{D}_i, \tag{1}$$

where $\mathcal{M} = \{\texttt{resnet18}, \texttt{resnet50}, \texttt{mobilenet_v2}, \texttt{efficientnet_b0}, \texttt{vit_tiny}\}$, $\mathcal{Q} = \{\texttt{fp32}, \texttt{fp16}, \texttt{int8_dynami}$
$\mathcal{B} = \{\texttt{pytorch_eager}, \texttt{torch_compile}, \texttt{onnxruntime}\}$, and $\mathcal{S} = \{1, 2, 4, 8, 16, 32\}$.

4.2 Constraints and Partial Invalidity

The deployment optimization problem is:

$$\begin{aligned} \boldsymbol{x}^* = \arg\max_{\boldsymbol{x} \in \mathcal{X}} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_j(\boldsymbol{x}) \le c_j, \quad j = 1, \ldots, J, \\ & \boldsymbol{x} \in \mathcal{X}_{\text{valid}} \subseteq \mathcal{X}. \end{aligned} \tag{2}$$

Here f denotes a deployment objective such as accuracy or throughput, g_j are constraint functions such as latency$_{p95}$ or peak memory, and $\mathcal{X}_{\text{valid}}$ denotes the unknown subset where evaluation does not crash.

Total constraint violation is:

$$V(\boldsymbol{x}) = \sum_{j=1}^{J} \max\left(0,\ g_j(\boldsymbol{x}) - c_j\right). \tag{3}$$

A configuration is feasible if $V(\boldsymbol{x}) = 0$ and $\boldsymbol{x} \in \mathcal{X}_{\text{valid}}$.

Definition 4.2 (Crash zone)**.** *The crash zone $\mathcal{X}_{crash} \subset \mathcal{X}$ is the subset of configurations for which evaluation fails to produce a valid result. The crash zone is not known a priori and must be discovered through evaluation.*

The effective search space decomposes as

$$\mathcal{X} = \underbrace{\mathcal{X}_{\text{crash}}}_{\text{crashed}} \cup \underbrace{(\mathcal{X}_{\text{valid}} \setminus \mathcal{C})}_{\text{valid but infeasible}} \cup \underbrace{\mathcal{F}}_{\text{feasible}}, \tag{4}$$

where $\mathcal{C} = \{\boldsymbol{x} : g_j(\boldsymbol{x}) \leq c_j,\ \forall j\}$ and $\mathcal{F} = \mathcal{X}_{\text{valid}} \cap \mathcal{C}$.

Definition 4.3 (Budget-constrained deployment optimization)**.** *Given $\mathcal{X}$, objective f, constraints $\{(g_j, c_j)\}_{j=1}^{J}$, and evaluation budget $B \in \mathbb{N}$, the task is to find the best feasible configuration using at most B evaluations, where each evaluation may crash and has variable cost $t(\boldsymbol{x}) > 0$.*

Two metrics capture optimizer performance. The first is simple regret:

$$r(n) = f(\boldsymbol{x}^*_{\text{global}}) - \max_{i \leq n,\ \boldsymbol{x}_i \in \mathcal{F}} f(\boldsymbol{x}_i). \tag{5}$$

The second is wasted budget fraction:

$$W(n) = \frac{|\{i \leq n : \boldsymbol{x}_i \notin \mathcal{F}\}|}{n}. \tag{6}$$

4.3 A Heuristic Characterization of Premature Exploitation

The central failure mode considered in this work is *premature exploitation*. In a crash-heavy space, an optimizer that transitions into model-guided sampling after only a small number of successful evaluations may have observed only a narrow subset of model families. Subsequent exploitation then reinforces that biased sample.

Remark 4.1 (Probabilistic intuition for premature exploitation)**.** *If the probability of sampling a specific model family in one trial is $p_m \approx 1/K$ for K families, and the survival probability is $(1 - \rho_{crash})$, then the probability of discovering that family in n_0 random trials is*

$$\mathbb{P}(\textit{discover family } m \textit{ in } n_0 \textit{ trials}) = 1 - (1 - p_m(1 - \rho_{crash}))^{n_0}. \tag{7}$$

For $K = 5$, $\rho_{crash} = 0.6$, and $n_0 = 10$, this yields approximately 0.57, which predicts that only about 4–6 of 10 seeds would discover a given family during the initial exploration stage. This is directionally consistent with the observed behavior of cold-start TPE in the edge-tight setting (Section 9).

5 Research Hypotheses

The proposed method is motivated by three hypotheses.

H1: Early structural coverage matters. In hierarchical deployment spaces, the main risk of early optimization is not merely poor local refinement but insufficient structural coverage. If early evaluations fail to touch promising model families or backend regimes, later model-guided search is conditioned on a biased subset of the space.

H2: Invalidity changes the exploration–exploitation trade-off. In standard black-box optimization, random or quasi-random early trials are often adequate to initialize a surrogate. In partially invalid spaces, the same number of trials may produce too few successful observations to support reliable model-guided exploitation.

H3: Warm-started exploitation is preferable to cold-start exploitation in hostile spaces. If an exploration phase can deliberately accumulate a more diverse set of valid and feasible observations, then a subsequent model-guided optimizer should perform better than if it had spent the same budget on unguided early exploration.

These hypotheses define the scientific scope of the paper. The contribution is therefore not framed as a universally superior optimizer, but as evidence for a specific architectural claim: in crash-heavy constrained deployment spaces, explicit feasible-first exploration improves the quality of subsequent model-guided search.

6 Method: Thermal Budget Annealing

6.1 Overview

TBA→TPE consists of two phases. Phase 1 uses simulated annealing to search for feasible regions while deliberately traversing structural choices. Phase 2 injects the full history from Phase 1 into a constrained TPE study and uses TPE for model-guided exploitation. The full optimizer is summarized in Algorithm 1.

6.2 Phase 1: Feasible-First Simulated Annealing

Phase 1 operates with a two-stage objective. Before any feasible configuration has been found, the optimizer minimizes total violation $V(\boldsymbol{x})$. Once a feasible configuration is found, the objective switches to maximizing $f(\boldsymbol{x})$ over feasible states.

The temperature follows an adaptive schedule with reheating:

$$T_{n+1} = \begin{cases} \alpha_s T_n & \text{if improved,} \\ T_n & \text{if not improved and patience not exceeded,} \\ \min(\beta T_n,\ \gamma T_0) & \text{if patience exceeded.} \end{cases} \tag{8}$$

In the reported experiments, $\alpha_s \in \{0.92, 0.95\}$ with slower cooling during the feasibility stage to maintain broader exploration, $\beta = 2.5$, $\gamma = 0.8$, and patience is 4 consecutive non-improving trials.

6.3 Acceptance Criteria

The acceptance rule depends on the current stage.

Feasibility stage. A new state is accepted deterministically if it reduces total violation. Otherwise, it is accepted with probability

$$\mathbb{P}(\text{accept}) = \exp\left(-\frac{V(\boldsymbol{x}') - V(\boldsymbol{x}_{\text{curr}})}{T_n \cdot \sigma_V}\right), \tag{9}$$

where

$$\sigma_V = \max\left(0.1,\ \frac{1}{J}\sum_{j=1}^{J} |c_j| \cdot 0.1\right). \tag{10}$$

Optimization stage. If the proposal is infeasible, it is rejected outright. If it improves the objective, it is accepted. Otherwise, it is accepted with probability

$$\mathbb{P}(\text{accept}) = \exp\left(-\frac{f(\boldsymbol{x}_{\text{curr}}) - f(\boldsymbol{x}')}{T_n \cdot \sigma_f}\right), \tag{11}$$

where

$$\sigma_f = \max\left(0.01,\ \frac{f_{\max} - f_{\min}}{2}\right). \tag{12}$$

Algorithm 1 TBA→TPE Hybrid Optimizer

Require: Search space $\mathcal{X}$, constraints $\{(g_j, c_j)\}$, objective f, budget B

$\mathcal{H} \leftarrow \emptyset$, $T \leftarrow T_0$, phase $\leftarrow$ `feasibility`
Sample n_{init} random configurations; evaluate and add to $\mathcal{H}$
$\boldsymbol{x}_{\text{curr}} \leftarrow$ best feasible in $\mathcal{H}$ (or least-violating if none feasible)
for $n = n_{\text{init}} + 1$ to adaptive handoff **do**
  Compute structural mutation probability $p_s(n)$
  if elite restart condition met and $n < 0.7B$ **then**
    $\boldsymbol{x}' \leftarrow$ 2nd or 3rd best feasible from $\mathcal{H}$
  else
    $\boldsymbol{x}' \leftarrow \text{ProposeNeighbor}(\boldsymbol{x}_{\text{curr}}, T, p_s(n))$
  end if
  Evaluate $\boldsymbol{x}'$; add $(\boldsymbol{x}', r')$ to $\mathcal{H}$
  if r' indicates crash **then**
    Update T as a non-improving step; keep $\boldsymbol{x}_{\text{curr}}$
  else if phase = `feasibility` **then**
    Apply violation-based Metropolis update
  else
    Apply objective-based Metropolis update with hard feasibility filter
  end if
  if $\text{HandoffReady}(\mathcal{H})$ **then**
    break
  end if
end for
Create Optuna study with constrained TPE sampler [Watanabe and Hutter, 2023]
for each $(\boldsymbol{x}_i, r_i) \in \mathcal{H}$ **do**
  Inject as completed trial with objective value and constraint metadata
end for
for $n = |\mathcal{H}| + 1$ to B **do**
  Request next configuration from TPE
  Evaluate and report result to the study
end for
return best feasible configuration observed

6.4 Hierarchical Neighbor Proposals

Neighborhood proposals must respect the hierarchical structure of the space. With probability $p_s(n)$, the optimizer performs a structural move by changing a categorical variable such as model family, backend, or quantization mode. Conditional variables activated by the new structure are sampled from their valid domains; deactivated variables are removed. With probability $1-p_s(n)$, the optimizer performs a numerical move by perturbing an active integer or continuous parameter.

The structural mutation probability decays linearly over the budget:

$$p_s(n) = 0.5 - 0.35 \cdot \frac{n - n_{\text{init}}}{B - n_{\text{init}}}. \tag{13}$$

This schedule reflects the intuition that early search should privilege structural exploration while later search should privilege local refinement.

6.5 Crash Handling and Exploration Mechanisms

Crashed configurations are never accepted as the current state. However, they are retained in history and therefore contribute to the optimizer's implicit map of invalid regions. This design allows the search to learn crash boundaries without accumulating in crash-prone states.

Two additional mechanisms stabilize exploration. *Elite restart* replaces the current state with the 2nd or 3rd best feasible configuration every 12 trials during the first 70% of budget. *Decaying snap-back* returns to the global best feasible state with probability $0.4(1 - n/B)$ after accepting a worse feasible move.

6.6 Adaptive Handoff into TPE

The handoff from exploration to exploitation occurs when the optimizer has observed sufficient evidence to initialize TPE reliably:

$$\text{handoff if } n \geq n_{\min}, \quad N_{\text{feas}} \geq 5, \quad N_{\text{bad}} \geq 3, \tag{14}$$

with a hard maximum of $n_{\max} = 15$ trials. Here N_{feas} is the number of feasible observations and N_{bad} is the number of crashed or infeasible observations. This criterion ensures that the TPE phase receives both positive and negative information about the space.

6.7 Phase 2: Warm-Started TPE

All Phase 1 evaluations are injected into an Optuna study as completed trials, including crashed and infeasible configurations together with constraint metadata. The constrained TPE acquisition can be expressed schematically as

$$\text{EI}_c(\boldsymbol{x}) = \frac{\ell(\boldsymbol{x})}{g(\boldsymbol{x})} \cdot \mathbb{P}(\boldsymbol{x} \in \mathcal{C} \mid \mathcal{H}), \tag{15}$$

where $\ell(\boldsymbol{x})$ and $g(\boldsymbol{x})$ are density estimates over good and bad observations, respectively. The crucial point is that Phase 2 begins from a broader and more informative set of initial observations than cold-start TPE would obtain from unguided early trials.

6.8 Computational Cost

The practical cost of deployment optimization is determined jointly by trial count and wall-clock time. Let $t(\boldsymbol{x})$ denote the time required to evaluate configuration $\boldsymbol{x}$. Then cumulative wall-clock cost is

$$T_{\text{wall}}(n) = \sum_{i=1}^{n} t(\boldsymbol{x}_i). \tag{16}$$

Although the current method does not explicitly optimize wall-clock cost, reducing wasted evaluations is valuable because crashes and infeasible trials may still consume substantial time through compilation, export, and repeated failed execution.

7 DeployBench: A Benchmark for Crash-Heavy Deployment Optimization

7.1 Design Principles

Existing HPO benchmarks [Eggensperger et al., 2021, Pfisterer et al., 2022] assume that most configurations produce valid results. DeployBench is designed to reflect deployment regimes where this assumption fails:

Table 1: Deployment search space variables.

Variable	Type	Domain	Condit
`model_name`	Categorical	{resnet18, resnet50, mobilenet_v2, efficientnet_b0, vit_tiny}	No
`backend`	Categorical	{pytorch_eager, torch_compile, onnxruntime}	No
`quantization`	Categorical	{fp32, fp16, int8_dynamic}	No
`batch_size`	Integer	$\{1, 2, 4, 8, 16, 32\}$	No
`num_threads`	Integer	$[1, 8]$	backend $\neq$ `tor`

Table 2: Deployment scenarios.

Scenario	Constraints	Objective	Budget
`edge_tight`	$\text{latency}_{p95} \leq 20$ms, memory ≤ 512MB	maximize accuracy	25
`server_throughput`	$\text{latency}_{p99} \leq 100$ms, memory ≤ 4096MB	maximize throughput	25

1. many random configurations are invalid (30–80% invalidity),
2. the space is hierarchical and mixed-variable,
3. constraints are hard rather than softly penalized,
4. evaluation costs vary substantially,
5. each trial is logged in JSONL format for reproducible behavioral analysis.

7.2 Synthetic Benchmarks

Two synthetic tasks were used during development and evaluation. *Crashy Branin* augments the standard Branin function with a categorical mode variable, an integer resolution variable, explicit crash zones, and feasibility constraints, yielding approximately 65% combined invalidity. *Hierarchical Rosenbrock* activates different subsets of continuous variables according to a categorical mode, thereby varying effective dimensionality across the space.

7.3 Real Deployment Benchmark

The real benchmark evaluates deployment configurations on ImageNette [Howard, 2020] using five pretrained vision models, three backends, three quantization modes, six batch sizes, and a conditional thread parameter. The resulting search space is summarized in Table 1.

Before accounting for conditional inactivity, this space contains 2160 structural combinations. Known crash patterns include `torch_compile` + `fp16`, `onnxruntime` with large models and large batch sizes, `int8_dynamic` + `vit_tiny`, and large model + large batch size causing GPU OOM. These patterns are intentionally not filtered from the search space.

Each evaluation follows a standardized procedure: model loading, optional quantization, backend compilation or export, 30 warmup iterations, 100 timed iterations, resource measurement, accuracy evaluation on 500 ImageNette validation images, and explicit memory cleanup. All evaluations are wrapped in exception handling so that a crash produces a logged result rather than terminating the experiment.

8 Experimental Setup

8.1 Baselines

Four baselines are evaluated under identical search spaces, budgets, and seeds:

1. **Random Search** [Bergstra and Bengio, 2012]: uniform random sampling, reporting the best feasible result.
2. **Optuna TPE** [Akiba et al., 2019, Watanabe and Hutter, 2023]: constrained tree-structured Parzen estimation with native conditional handling.
3. **Constrained Bayesian Optimization**: Gaussian process with constrained expected improvement [Gardner et al., 2014, Gelbart et al., 2014].
4. **TBA (pure SA)**: the feasible-first exploration stage without TPE handoff. This ablation isolates the contribution of warm-starting.

8.2 Metrics

Performance is assessed along four dimensions:

1. best feasible objective (mean $\pm$ std across seeds),
2. wasted budget fraction $W(B)$,
3. time to first feasible configuration,
4. model-family discovery rate.

8.3 Hardware, Software, and Protocol

All experiments were conducted on an NVIDIA RTX 5080 Laptop GPU (16 GB GDDR7) with an Intel Core Ultra 9 275HX and 16 GB DDR5 RAM. Software versions: PyTorch 2.10.0 (CUDA 12.8), torchvision 0.25.0, ONNX Runtime 1.24.4, Optuna 4.2, Python 3.14. All experiments use 10 random seeds (0–9). Conditional parameters are exposed consistently across methods, and crashed trials are recorded rather than silently filtered.

9 Results

9.1 Synthetic Benchmarks: Budget Sweep

Table 3 reports results on Crashy Branin across budgets from 10 to 50.

Three findings emerge from Table 3 and Figures 1–2. First, at $B = 20$, the hybrid outperforms all baselines including TPE (-3.18 vs -4.19), indicating that warm-starting with crash-aware exploration data can outperform cold-start model-guided search when the budget is tight. Second, at $B = 25$–30, the hybrid and TPE achieve comparable objectives, but the hybrid exhibits lower variance (± 0.79 vs ± 1.33 at $B = 25$), indicating more consistent performance. Third, the wasted budget fraction for the hybrid (23–34%) is consistently lower than TPE's (29–54%), with the largest gap at small budgets. This pattern is consistent with H2 and H3: the main gain comes from better early exploration rather than stronger asymptotic exploitation.

9.2 Ablation

The ablation indicates that restricting restarts to the first 70% of budget is the most impactful component among the pure-TBA modifications, while the TPE handoff provides the largest overall gain over pure SA with substantially reduced variance.

Table 3: Crashy Branin: best feasible objective (higher is better) and wasted budget fraction. Bold indicates best objective at each budget.

	TBA→TPE		TBA (SA)		TPE		Random	
B	Obj	W	Obj	W	Obj	W	Obj	W
10	−5.83	34%	**−5.65**	40%	−7.29	54%	−8.44	65%
15	−5.54	25%	−4.71	32%	**−2.59**	42%	−6.37	66%
20	**−3.18**	30%	−4.62	27%	−4.19	40%	−5.17	63%
25	−2.29	28%	−3.68	28%	**−2.14**	35%	−4.12	64%
30	−2.05	30%	−2.93	27%	**−1.97**	31%	−4.41	67%
40	−1.57	26%	−3.95	24%	**−1.38**	30%	−2.98	63%
50	−1.45	23%	−2.66	20%	**−1.15**	29%	−2.56	66%

Table 4: Ablation on Crashy Branin at $B = 30$.

Variant	**Best Objective**	**Wasted %**
TBA full (SA + all components)	-2.93 ± 2.37	27%
w/o no-late-restarts	-3.62 ± 2.58	30%
w/o decaying snap-back	-3.45 ± 2.71	29%
w/o adaptive p_s	-3.21 ± 2.50	28%
TBA→TPE hybrid	$\mathbf{-2.05 \pm 0.71}$	30%

9.3 Real Deployment: Edge-Tight Scenario

Table 5 reports results under tight edge deployment constraints.

The aggregate accuracies are close, and the hybrid's edge-tight mean accuracy (0.761) is essentially tied with random search (0.762). The hybrid's advantage is not a large mean-accuracy gain but rather improved discovery behavior and budget efficiency. The more informative result is model-family discovery. Table 6 shows that cold-start TPE discovers `vit_tiny`, the best constrained model family in this setting (76.6% accuracy), in only 4 out of 10 seeds. In the remaining seeds, TPE converges to `resnet50` (accuracy 0.746) and exploits it without evaluating `vit_tiny`. This is the premature exploitation phenomenon described in Remark 4.1.

This is the central empirical finding. The hybrid avoids the `resnet50` trap because Phase 1 explores broadly enough to encounter `vit_tiny` before the exploitation phase narrows the search. Random search also discovers `vit_tiny` frequently, but it wastes 77% of its budget on invalid trials, nearly double the hybrid's 41%.

The per-seed model discovery table (Table 7) reveals the behavioral difference in detail. Several TPE seeds converge to `resnet50` without ever evaluating `vit_tiny`, while the hybrid's structural exploration during Phase 1 reaches `vit_tiny` before exploitation begins.

9.4 Real Deployment: Server-Throughput Scenario

Under relaxed constraints, TPE outperforms the hybrid on throughput (Figure 5). This is expected: with fewer configurations crashing in the relaxed regime, TPE's cold-start exploration is less hindered, and its model-guided sampling becomes sufficient. This result is important because it clarifies the regime boundary: feasible-first exploration provides the largest benefit when the space is hostile.

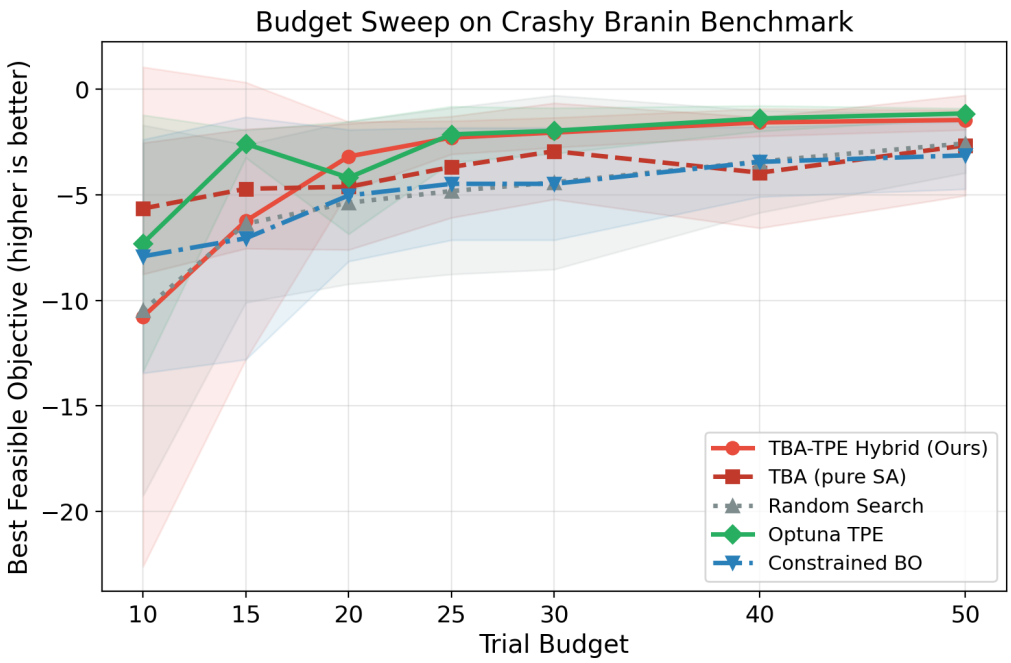

Figure 1: Best feasible objective versus evaluation budget on Crashy Branin (10 seeds, mean ± std). The hybrid outperforms cold-start TPE at $B = 20$ and tracks it closely at higher budgets, while pure SA and random search lag behind.

Table 5: Edge-tight deployment: latency$_{p95} \leq 20$ms, memory ≤ 512MB, budget=25, 10 seeds.

Optimizer	**Accuracy (mean ± std)**	**Wasted %**	**Feasible seeds**
Random Search	0.762 ± 0.008	77%	10/10
TBA→TPE Hybrid	0.761 ± 0.010	41%	10/10
TBA (pure SA)	0.755 ± 0.015	33%	10/10
Optuna TPE	0.754 ± 0.010	35%	10/10

10 Analysis

10.1 Why Does TPE Miss the Best Model?

TPE's failure to discover `vit_tiny` in 6/10 edge-tight seeds can be explained by the interaction between crash rate and exploration depth. TPE uses an initial random design before switching to model-guided sampling. In a space where 65–80% of configurations crash or violate constraints, those initial trials may yield only 2–4 successful evaluations, potentially all from the same model family. Once the density estimator is fitted to this narrow sample, exploitation reinforces the same family rather than expanding structural coverage.

Using the formula from Remark 4.1 with $K = 5$, $\rho_{\text{crash}} = 0.6$, and $n_0 = 10$: $\mathbb{P} \approx 0.57$. This predicts that roughly 4–6 out of 10 seeds would discover any particular family, closely matching the observed 4/10 rate.

By contrast, the hybrid uses Phase 1 to widen structural coverage before handing off to TPE. This does not guarantee discovery of every promising family, but it raises the probability that the best family enters the candidate set prior to exploitation.

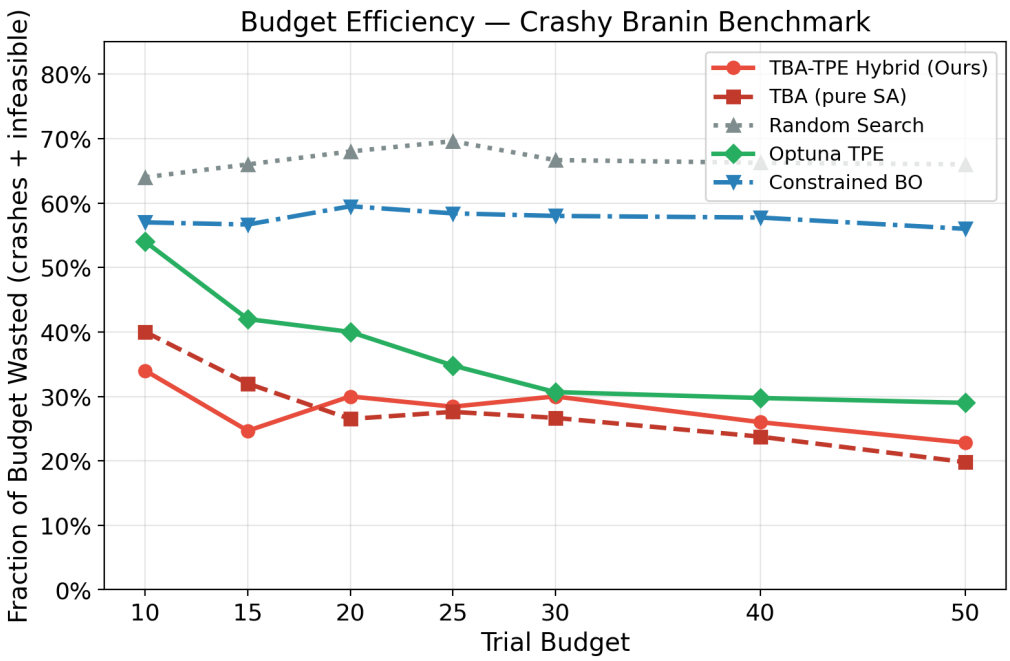

Figure 2: Wasted budget fraction versus evaluation budget on Crashy Branin. The hybrid consistently wastes less budget than TPE across all budgets tested. Random search wastes 63–67% regardless of budget.

Table 6: Seeds (out of 10) discovering `vit_tiny`, the globally best architecture under edge-tight constraints.

Optimizer	**Discovery rate**
TBA→TPE Hybrid	**8 / 10**
Random Search	8 / 10
TBA (pure SA)	7 / 10
Optuna TPE	4 / 10

10.2 Random Search as a Diagnostic

An important observation is that random search matches the hybrid's `vit_tiny` discovery rate (8/10) while wasting far more budget. This is diagnostically useful. It shows that `vit_tiny` is not hidden in an obscure corner of the space. The problem is therefore not reachability but *search dynamics*: TPE's model-guided policy steers away from that family after committing early to `resnet50`. The hybrid preserves much of the structural coverage of broad exploration while avoiding the extreme inefficiency of random search.

10.3 Exploration–Exploitation Boundary

The contrast between edge-tight and server-throughput results suggests a practical boundary. When invalidity is high enough that early trial budgets are dominated by crashes and infeasible configurations, warm-starting from crash-aware exploration is beneficial. When invalidity is lower, cold-start TPE is already able to obtain sufficient successful observations for effective model-guided search.

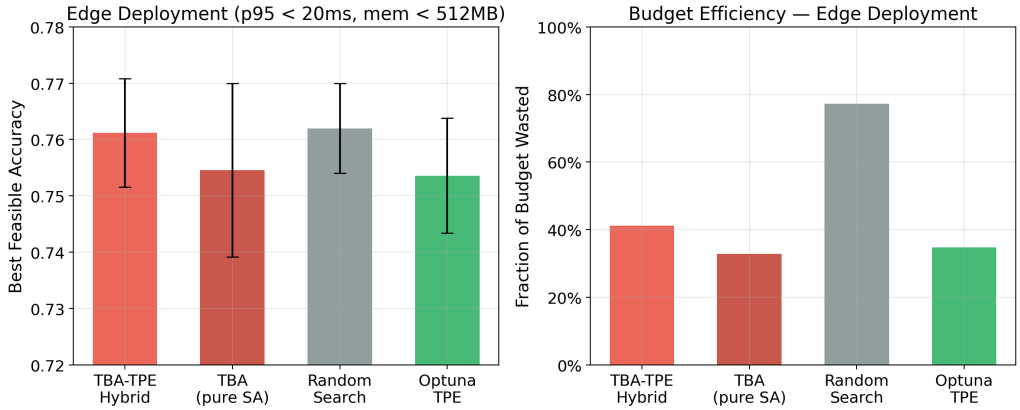

Figure 3: Edge-tight deployment results: accuracy and wasted budget fraction by optimizer. The hybrid achieves accuracy comparable to random search while wasting roughly half the budget.

Table 7: Best model family discovered per seed under edge-tight constraints. `vit` = ViT-Tiny (0.766), `r50` = ResNet-50 (0.746), `r18` = ResNet-18 (0.724), `eff` = EfficientNet-B0 (0.742).

Seed	TBA→TPE	TBA (SA)	Random	TPE
0	vit	vit	vit	r50
1	vit	vit	vit	vit
2	vit	r50	vit	r50
3	vit	vit	vit	vit
4	eff	vit	r50	r50
5	vit	vit	vit	r50
6	vit	vit	vit	vit
7	vit	r50	vit	r50
8	eff	r50	r18	vit
9	vit	vit	vit	r50
vit rate	8/10	7/10	8/10	4/10

11 Discussion

The results support the paper's central claim with appropriate nuance. The hybrid is not universally better than TPE. Rather, it changes the exploration–exploitation trade-off in settings where the early trial budget is too small, and the invalidity rate too high, for cold-start model-guided search to form a sufficiently broad view of the space.

H1 (structural coverage) is supported by the discovery-rate analysis: the hybrid's structural mutations during Phase 1 produce broader model-family coverage than TPE's cold-start random sampling in crash-heavy spaces. However, the effect size is moderate (8/10 vs 4/10), and the hybrid also misses `vit_tiny` in 2 seeds.

H2 (invalidity changes the trade-off) is supported by the contrasting results across scenarios: the hybrid helps substantially under edge-tight constraints but provides marginal benefit under server-throughput constraints.

H3 (warm-start helps) is supported by the ablation: TBA→TPE achieves -2.05 ± 0.71 versus TBA pure SA's -2.93 ± 2.37 at $B = 30$, with both better mean and substantially lower variance.

A secondary implication concerns evaluation metrics. In deployment optimization, best feasible objective alone is insufficient. The hybrid's advantage manifests primarily in budget efficiency and discovery consistency, not in mean objective superiority. Trial waste is operationally

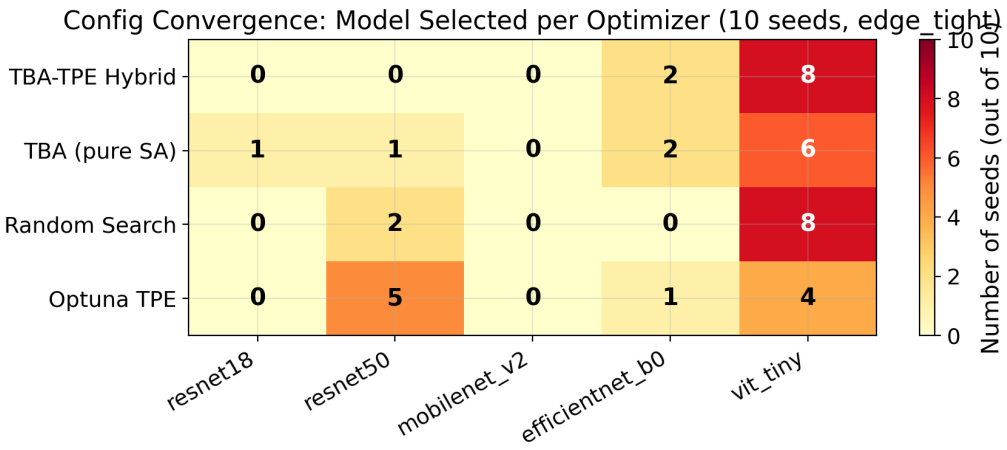

Figure 4: Model family convergence per seed under edge-tight constraints. Each cell shows which model family each optimizer converged to. TPE locks onto `resnet50` in 6 of 10 seeds, while the hybrid discovers `vit_tiny` in 8 of 10 seeds.

Table 8: Server-throughput deployment: latency$_{p99} \leq$ 100ms, memory $\leq$ 4096MB, budget=25, 10 seeds.

Optimizer	**Throughput QPS (mean $\pm$ std)**	**Wasted %**	**Feasible seeds**
Optuna TPE	5423 ± 1325	24%	10/10
TBA$\rightarrow$TPE Hybrid	5087 ± 1787	23%	10/10
Random Search	4758 ± 1399	54%	10/10
TBA (pure SA)	4131 ± 2036	18%	10/10

meaningful because each evaluation may require model export, runtime compilation, repeated timing, and memory measurement. A method that obtains comparable feasible performance with substantially lower waste can be preferable in practice.

12 Limitations and Threats to Validity

Several factors limit the scope of the present conclusions.

Single hardware target. All experiments use a single GPU and software stack. Backend behavior, compiler stability, and quantization support may differ on other systems.

Single deployment domain. The benchmark focuses on image classification deployment. Extensions to NLP serving, speech inference, and multi-model pipelines [Zaharia et al., 2024] remain future work.

Benchmark scale. The deployment space contains roughly 2000 structural combinations. Larger spaces may amplify the premature exploitation effect, but they may also challenge the exploration capacity of Phase 1.

Statistical scope. Ten seeds are sufficient for an initial empirical study but limit precision. Larger seed counts and additional hardware targets would strengthen the conclusions.

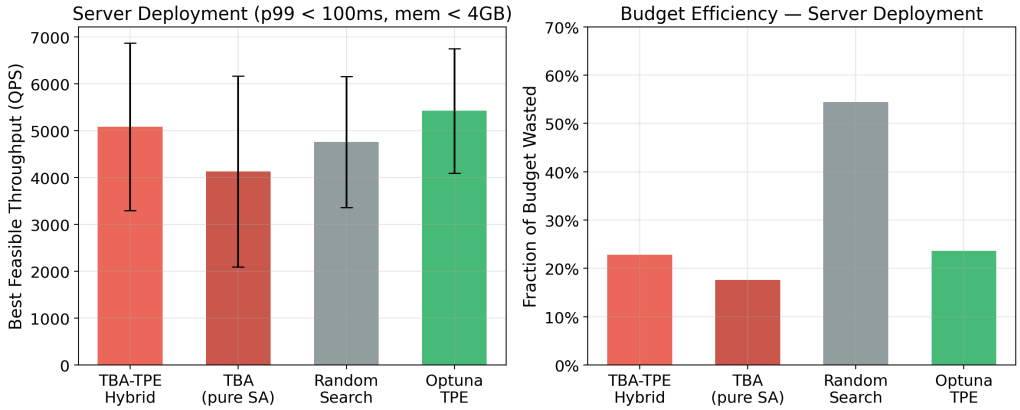

Figure 5: Server-throughput deployment results: throughput (QPS) and wasted budget by optimizer. Under relaxed constraints, TPE achieves the highest throughput, confirming that the hybrid's advantage is specific to crash-heavy regimes.

Synthetic–real gap. The synthetic benchmarks reproduce structural invalidity and hidden crash regions but do not replicate every detail of real deployment failures.

Aggregate accuracy parity. The hybrid's edge-tight mean accuracy (0.761) is essentially tied with random search (0.762). Its advantage manifests primarily in discovery behavior and budget efficiency rather than large gains in average objective.

13 Conclusion

This paper studied deployment optimization as a budget-constrained search problem in a partially invalid hierarchical space. The main result is not that simulated annealing outperforms mature model-guided optimizers in general. Instead, the evidence supports a narrower and more useful conclusion: when invalidity is high and budgets are small, *the quality of early exploration determines the effectiveness of later exploitation*. Warm-starting TPE with feasible-first exploration improves model-family discovery under tight constraints and reduces wasted budget relative to cold-start TPE.

Three directions appear especially promising for future work. First, the handoff criterion could be made adaptive to observed structural diversity rather than simple counts. Second, hardware-transfer experiments could test whether feasibility maps learned on one device reduce search cost on another. Third, the benchmark can be extended beyond image classification to multi-model and language-model deployment scenarios.

The benchmark suite and optimizer implementation are available at https://github.com/Chrislysen/Constrained-ML-Deployment.

A Additional Method Detail

A.1 Adaptive Structural Mutation Schedule

The structural mutation probability is decayed linearly:

$$p_s(n) = 0.5 - 0.35 \cdot \frac{n - n_{\text{init}}}{B - n_{\text{init}}}, \tag{17}$$

where n_{init} is the number of initial random trials.

A.2 Snap-Back and Elite Restart

The pure-TBA variant uses two stabilization mechanisms: (i) elite restart to the 2nd or 3rd best feasible state during early search, and (ii) decaying snap-back to the best feasible state after accepting a worse move. These mechanisms were retained because they improved the ablation metrics on Crashy Branin (Table 4).

B Computational Cost

Table 9: Wall-clock time for complete experimental evaluation.

Experiment	**Wall-clock**
Synthetic budget sweep (5 opt. $\times$ 7 budgets $\times$ 10 seeds)	$\sim$15 min
Edge-tight deployment (4 opt. $\times$ 10 seeds $\times$ 25 trials)	$\sim$90 min
Server-throughput (4 opt. $\times$ 10 seeds $\times$ 25 trials)	$\sim$90 min
Total	$\sim$3.5 hours

All experiments were conducted on a single NVIDIA RTX 5080 laptop GPU. No multi-GPU or cloud compute was required.

C Reproducibility Checklist

1. All deployment experiments were executed on a single RTX 5080 Laptop GPU with the software versions listed in Section 8.
2. Each optimizer was evaluated on 10 random seeds (0–9).
3. Crashed trials were logged rather than discarded.
4. The search-space specification, benchmark definitions, and experiment scripts are available in the public repository.
5. The reported tables are generated from saved JSONL result files.
6. Conditional parameters are exposed identically to all optimizers.

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